

Shortest Paths

What are Shortest Paths?

Given a directed, weighted graph $G = (V, E)$ and a source vertex s , the **shortest path problem** aims to find the path with the minimum total weight (or length) from s to all other vertices in G .

The weight of a path is the sum of the weights of its constituent edges.

Shortest Path Tree

The collection of all shortest paths from a single source s to all other reachable vertices forms a **shortest path tree**. This tree is rooted at s , and every path from s to any other vertex v in the tree is a shortest path in G .

Assumptions

For simplicity, when discussing shortest path algorithms, we often assume: - All vertices are reachable from the source vertex s . - Consequently, a shortest path to each vertex always exists.

Subpath Property

Any subpath of a shortest path is itself a shortest path.

Proof: Consider a shortest path P from s to t . Let P be composed of three segments: P_1 from s to u , P_2 from u to v , and P_3 from v to t . So, $P = P_1 \rightarrow P_2 \rightarrow P_3$. Assume, for contradiction, that P_2 is not a shortest path from u to v . This means there exists a shorter path, say Q_2 , from u to v . If we replace P_2 with Q_2 in the original path P , we would form a new path $P' = P_1 \rightarrow Q_2 \rightarrow P_3$. The total weight of P' would be $w(P_1) + w(Q_2) + w(P_3)$. Since $w(Q_2) < w(P_2)$, it follows that $w(P') < w(P)$. This contradicts our initial assumption that P was a shortest path from s to t . Therefore, P_2 must be a shortest path from u to v . This logic applies to any subpath.

Types of Shortest Path Problems

Shortest path problems can be categorized based on various factors:

Category	Description
Source/Target	Single source (to all others), Single source-single target, All-pairs (between all pairs of vertices)
Edge Weights	Non-negative, Arbitrary (can be negative), Euclidean distances

Category	Description
Cycles	No cycles (DAGs), No negative cycles (general graphs), Cycles allowed (but no negative cycles for most algorithms)

Related

- [[Graphs]]
- [[Dijkstra's Algorithm]]
- [[Shortest Paths on DAGS]]